Scaling from Metawebs to Realised Webs: A Hierarchical Approach to Network Ecology

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Abstract: Ecological networks provide a critical framework for understanding the architecture of biodiversity and predicting ecosystem responses to environmental change. However, the application of network ecology is often hindered by a lack of clarity regarding the assumptions inherent in different network representations. Here, we present a hierarchical framework that distinguishes between 'metawebs' (representing the fundamental feasibility of interactions) and 'realised webs' (representing interactions expressed in specific spatiotemporal contexts). We contrast our conceptual approach with recent data-centric reviews, focusing instead on the theoretical gradients that govern network construction. We identify five core processes that drive the transition from potential to realised interactions: evolutionary compatibility and co-occurrence, which define the feasibility of links; and abundance, diet choice, and non-trophic interactions, which determine their realisation. Furthermore, we map these processes onto a methodological spectrum of network construction, distinguishing between inductive approaches (*e.g.,* trait-matching and stochastic models) that infer structure from observation, and deductive approaches (*e.g.,* neutral and optimal foraging models) that generate structure from mechanistic first principles. By making explicit the assumptions and scale-dependent processes underpinning these different representations, this framework clarifies the scope of inference possible with each approach, ultimately facilitating more robust predictions of biodiversity dynamics in the anthropocene.

Keywords: food web, network construction, biodiversity, scale and process, interaction modelling, trophic network

1 Introduction

At the heart of modern biodiversity science are a set of concepts and theories about species richness, stability, and function, which have been discussed since the foundational work of [1], [2], and [3] and more recently [4]. These relate to the abundance, distribution, functions, and services that biodiversity provides. Network representations of interactions among organisms are increasingly argued to be an asset to understanding and predicting the impacts of multiple, simultaneous stress on biodiversity (*e.g.,* foundational studies on food web structure and robustness: [5–8]). Documenting interactions is thus one of the fundamental building blocks of community ecology and provides a powerful abstraction for mathematical and statistical modelling of biodiversity to make predictions, and to mitigate and manage threats [9].

However, there is a growing discourse around limitations to the interpretation and applied use of networks, which have been recognised since early discussions of sampling effort and aggregation in food webs ([5,10,11]; recent discussions: [12,13]). Against this, it is important to evaluate the value and the limitations of the various network conceptualisations and how these relate to biodiversity concepts, such as community structure or ecosystem function [14]. In this perspective we aim to provide an overview of different **food web** representations, particularly how each representation embeds assumptions about the processes that determine interactions (Section 3) about the levels of organization at which this occurs (*i.e.* the biological, ecological, spatial/temporal scales) and the way in which we construct the resulting networks (Section 4).

Network construction reflects both the data used and the theories governing species interactions. We still lack a clear explanation of the different assumptions and scale dependent processes that underpin network construction alongside extensive discussions about the challenges relating to data collection and observation [6,10,14–20]. Such an understanding should deliver an acceleration in capacity to more effectively predict the impact of multiple stressors on biodiverse communities.

In their recent work, [21] synthesised ecological networks based on node resolution and link type, with a focus on the data and methodologies underlying each representation. Here, we take a complementary theory-driven perspective by framing network construction as a hierarchical transition from feasibility (metawebs) to realisation. In the following sections, we review how nodes, edges, and interaction processes shape different network representations, highlighting how changes in assumptions alter ecological scale, node and link resolution, modelling approaches, and the relative importance of evolutionary versus ecological processes. We conclude by aligning these representations with key questions in biodiversity science in the Anthropocene.

2 Setting the Scene: The Not So Basics of Nodes and Edges

Ecological networks serve multiple uses, representing an 'object' from which inferences can be made. While many aspects of community structure can be analysed without networks (*e.g.,* through trait distributions or abundance patterns) networks provide a formal framework for capturing the organisation of interactions among species. The study of network structure and topology has a long history in ecology, rooted in early theory on energy flow [2,22], and later extended to questions of robustness, stability, and complexity *e.g.,* [7,20,23–25]. More recent work has built on this foundation to link network structure to ecosystem functioning, persistence, and dynamical behaviour *e.g.,*[26–28]. Networks are therefore commonly treated as response variables in tests of ecological theory and statistical models of the generative processes that give rise to observed structure, and are widely used to compare communities across environmental gradients or through time [*e.g.,* 29,30]. They also provide a platform for evaluating downstream responses to perturbations, including secondary extinctions and robustness to species loss [*e.g.,* 7,31–33], as well as for implementing dynamical process that inference about stability, ecosystem function, invasions, climate change, contaminants, and extinction cascades [*e.g.,* 34,35,36]. Against this backdrop of multiple research agendas, the definition of 'edges' and 'nodes', and the levels of organisation at which they are defined, take many forms [17,37], each of which encode a series of assumptions within a network. Here we introduce a perspective on these baseline assumptions.

2.1 How do we define a node?

While nodes are conventionally described as representing species, in practice they may correspond to a range of taxonomic and non-taxonomic units, including sub-species, genera, or families, as well as trophic species (*e.g.,* [38,39]), feeding guilds (*e.g.,* [40]), or life-stage–specific subsets of species (*e.g.,* [41]). These choices reflect differences in the level, type, and consistency of resolution at which networks are constructed, rather than aggregation per se. In many cases, aggregation is applied explicitly on top of more highly resolved—or unevenly resolved—data to address particular questions, as in the construction of trophic species, and the underlying data may remain available for alternative analyses. Nevertheless, representing nodes at coarser or mixed resolutions can limit taxon-specific inference (*e.g.,* whether species *a* consumes species *b*), bias estimates of degree distributions—particularly generality and vulnerability—and complicate downstream analyses such as extinction or invasion dynamics, where species identity and the consequences of loss may be obscured [41,42]. At the same time, there are clear justifications for using aggregated representations when the distribution of interactions among functional or trophic units is more informative than species-level detail, for example when analysing extinction patterns across feeding guilds [43]. More broadly, issues of resolution, scale, and sampling have long been recognised as central to the construction and interpretation of food webs (*e.g.,* [6,44]).

2.2 What is captured by an edge?

Understanding edges requires distinguishing between *potential* and *realised* links - potential links reflect feasibility, whereas realised links reflect fluxes such as energy transfer. Links within food webs are thus a representation of either potential links between species [45] or fluxes within a system *e.g.,* energy transfer or material flow as the result of the feeding links between species [2,46]. Edges can thus correspond to different 'currencies' [21]. There are also a myriad of ways in which the links themselves can be specified. Links between species can be treated as present or absent (*i.e.,* binary), may be defined as probabilities [47,48] or by continuous functions which further quantify the strength of an interaction [49]. Link definition depends on both the ecological currency and how interactions are represented. For example, feasibility is unlikely to accommodate flux, but does align with binary or probability representations. Taking a food web that consists of links representing feasible interactions among a collection of species will be meaningless if one is interested in understanding the flow of energy through the network as the links are not environmentally/energetically constrained.

2.3 Network representations

Given these definitions of nodes and edges, ecological networks can be divided into two broad types: **metawebs**, which represent all *potential* interactions within a species pool [44], and **realised networks**, which represent the subset of interactions expressed within a particular community at a given place and time. These representations differ in both the scales at which they are constructed and the processes assumed to generate their structure.

A metaweb is fundamentally a list of *feasible* interactions between species. Feasibility is determined by trait complementarity, typically related to feeding, and can be further constrained by species co-occurrence, producing a transition from *global* to *regional* metawebs. Metawebs therefore identify evolutionarily and regionally plausible interactions, ecologically impossible (*i.e.,* forbidden) links [50], and the potential diet breadth of species [51].

In contrast, realised networks are localised in space and time, with links shaped by co-occurrence, environmental conditions, and diet choice. Even when represented as binary matrices, their links implicitly reflect interaction strength and energy transfer. Realised networks are therefore not simple spatial or temporal subsets of metawebs; they emerge from the processes governing whether feasible interactions are expressed. Consequently, a metaweb and realised network containing the same species may differ substantially in structure because link presence is governed by different constraints. Links absent from a metaweb represent infeasible interactions, whereas links absent from a realised network reflect context-dependent ecological constraints.

3 From Nodes and Edges to Process and Constraints

In the previous section we discussed how the definition of nodes and edges, representing different scales and processes, lead to the concept of a metaweb and a realised web. The fundamental take-homes are that nodes vary in their resolution, edges vary in what kind of process they represent and the intersection of these, defined by meta- vs. realised webs, underpins distinct lines of inquiry and constraints on the type of inference we can make with networks. Here we reveal five core constraints across evolutionary and ecological scales that further delineate the transition from meta- to realised webs, exposing processes that determine the nature of links among nodes: evolutionary compatibility, co-occurrence, abundance, diet choice, and non-trophic interactions Figure 1.

[Figure 1 about here.]

3.1 Processes that determine the feasibility of an interaction

Evolutionary compatibility and co-occurrence are the two principal processes that 'act' at the species pair of interest and define feasibility. The scale of inference and set of processes embodied in these two constraints typically combine to define a 'list' of interactions that are viable/feasible and defined strictly as present/absent. Reflecting on the previous section, nodes are typically species and rules defining edges are defined by trait complementarity (phylogenetic) and/or co-occurrence. Here we provide more insight into each process.

Evolutionary compatibility

This constraint is defined by shared (co)evolutionary history between consumers and resources [52–55] which is manifested as 'trait complementarity' between two species [56]. In this body of theory, the consumer has the 'correct' set of traits that allow it to acquire and consume the resource. Interactions that are not compatible are defined as forbidden links [50]; *i.e.,* they are not physically possible and will *always* be absent within a network.

Networks do not properly arise from models based on this constraint. Instead, interacting species pairs are defined and these are represented as binary (possible vs forbidden) or probabilistic [47]. For example, in the metaweb constructed by [57] probabilities are quantified as the confidence of a specific interaction being *possible* between two species. A network constructed on the basis of evolutionary compatibility is conceptually aligned with a 'global metaweb', and gives us information as to the global feasibility of links between species pairs despite the fact that they do not co-occur (see Figure 1).

(Co)occurrence

The co-occurrence of species in both time and space is a fundamental requirement for an interaction between two species to occur (at least in terms of feeding links). Although co-occurrence data alone is insufficient for building an accurate and ecologically meaningful representation of *feeding links* [58], it is still a critical process that determines the possible realisation of a feeding. Knowledge on the co-occurrence of species allows us to spatially constrain a global metaweb to reflect regional metawebs [59]. In the context of Figure 1 this would be the metawebs for regions one and two.

We reinforce that these two constraints don't deliver a network *per se,* but a list of feasible species pairs. Although it is possible to build a network from the list of interactions generated by these constraints, it is important to be aware that the structure of this network is not constrained by any community context - just because species are able to interact does not mean that they will [60,61].

3.2 Processes that realise networks

In contrast to the above, here we highlight three processes that influence the *realisation* of an interaction between species and thus form the conceptual basis for realised networks. As we show in Figure 1, a 'truly realised' network is the product of properties of the community (**abundance** and **non-trophic interactions**) and the individual (**diet choice**). This represents a conceptual shift from considering the feasibility for species pairwise interactions to considering the edge as a representation of energy flow. Such a transition requires information about how the community, the environment and the individual *constrains* network topology as defined by consumer choice ([62], Section 2.3)

Abundance

Abundance as a realising process emerges from a null model for energy acquisition: organisms feeding randomly will consume resources in proportion to their abundance [63]. Here, abundance of different prey species influences the distribution of links in a network [64] by defining a preference linked to individuals among species meeting [47,60]. Abundance data (linked to a derived metaweb) delivers a foundation ruleset that can define the distribution and strength of links. Of note, however, is that such abundance constrained interactions are not necessarily contingent on there being any compatibility between species [65–67].

Diet choice

It is well established that consumers make more active decisions than eating items in proportion to their abundance [63]. Ultimately, consumer choice is underpinned by an energetic cost-benefit framework centered around profitability and defined by traits associated with acquisition and consumption of a resource [68,69]. Energetic constraints are invoked to construct networks in a myriad of ways [*e.g.,* 42,70–72].

Unlike metaweb approaches, these models generate realised webs as emergent outcomes of consumer behaviour. We also here make a distinction, developed below, with models like the Niche Model [39], where diet choice is implicit in its probabilistic network generating function, but it is working to replicate the *expected* structure of the network, and this structure does not emerge from node-based rules. Note that we select diet choice as a term to capture rules linked to optimal foraging [73] and metabolic theory [74]; it is a sensible ‘umbrella concept’ for capturing the energetic constraint on of the distribution and strength of interactions.

Non-trophic interactions

We include non-trophic interactions [see 75] here not as a determinant of links, but a modifier of them - they are the community context above and beyond co-occurrence and abundance. Non-trophic interactions include competition for space, predator interference, refuge provisioning, recruitment facilitation as well as non-trophic effects that increase or decrease mortality. These interactions specifically modify either the realisation or strength of trophic interactions [28,31,76–78] and represent direct (*e.g.,* predator a outcompetes predator b) and indirect (*e.g.,* mutualistic/facilitative interactions) mechanisms.

Some interactions, such as pollination, occupy an intermediate position in this framework, as they combine trophic components (*e.g.,* resource consumption) with non-trophic effects that influence reproduction, recruitment, and population persistence [79–81]. They operate on the realisation of a network by altering the fine-scale distribution and abundance of species and relative contributions of direct and indirect effects to biomass, persistence, stability and the functioning of the communities [75,82–84].

4 Network construction

The above five processes are central to understanding the assumptions inherent in building different types of networks. Each of the processes, or combinations thereof, deliver a unique set of boundary conditions on what a network represents and can be used for. Here we build on the introduction of these five processes to further categorise the approaches to constructing networks. In doing so also introduce more detail on a variety of methodologies used to construct networks.

4.1 Why construct networks?

Networks are a representation of biodiversity. In a perfect world, we might know about all interactions. However, the empirical collection of interaction data is both costly and challenging to execute [50,85,86]. In the absence of robust empirical data, we construct models that facilitate interpolation and gap-filling of existing empirical datasets [*e.g.,* 87,88–90], predict the feasibility of interaction among pairs of species, or directly predict network structure [see 91 for a broader discussion].

They are unique in delivering more than just estimates of species richness. As noted in the introduction, a network embodies the organising structure of biodiversity and allows numerous opportunities for 'downstream' analysis, including the comparison of structures, estimation of energy flux or extinction dynamics and ultimately form the structural inputs to dynamical systems models that facilitate ecological and conservation relevant inference about productivity-diversity-stability-function relationships [27] in space and time. But making such inferences requires careful attention to one or more of the processes discussed in Section 3. While these network representations may be simplifications of the truth [92], they remain critically useful tools as they allow us to move beyond simple descriptions of species richness to test hypotheses about community architecture and ecosystem functioning that would be otherwise impossible to assess.

4.2 Construction through induction

Constructing feasible or realised networks can be framed as an 'inductive reasoning' process where insight and generalisation arises from a set of observations and relationships. Inductive reasoning as a foundation for network construction is implemented at node and network levels. When applied at the node level, species-specific networks are created and judged by their association with expected feeding interactions. When applied at the network level, networks are judged by their structural properties.

4.2.1 Species specific networks: construction through node level induction

Constructing feasible networks and facilitating the interpolation or gap-filling of existing empirical datasets on sets of species interactions can be framed as an 'inductive reasoning' process where insight and generalisation arises from a set of observations and relationships about feeding. All methods in this inference space rest on a set of three assumptions: there are a set of 'feeding rules' that underpin interaction feasibility [93]; these rules are phylogenetically conserved [54,94]; and they can be specified by matching the traits between consumer and resource.

Evolutionary compatibility and co-occurrence constraints have been critical to the construction of 'first draft' networks for communities for which we have no interaction data [57]. They are also central to interpolation in data poor regions and predicting interactions for 'unobservable' communities *e.g.,* prehistoric networks [43,95,96] or future, novel community assemblages [97]. Furthermore, they have the capacity to evaluate a role of interactions among species relative to their distribution by accounting for the role of the environment and the role of species interactions [98–101]. There are substantial data requirements for these approaches including expert knowledge, species traits and phylogenetic relationships and/or interaction data on related species or communities.

Feeding rules can be specified in several ways. Expert-based approaches define feasible interactions using trait matching [102,103], while mechanistic approaches often rely on body-size relationships between consumers and resources [104,105]. Alternatively, statistical and machine-learning methods infer interaction probabilities from observed networks using ecological predictors such as traits and phylogeny [106].

Rules are also defined by correlating real world interaction data with suitable ecological proxies for which data is more widely available (*e.g.,* traits) using some sort of binary classifier (see [106] for an overview). These include generalised linear models [*e.g.,* 107], random forest [*e.g.,* 108], trait-based k-NN [*e.g.,* 109], and Bayesian models [*e.g.,* 110,111].

Finally, graph embedding uses the structural features of a known network to infer the position of species in an unknown network through the decomposition of the interaction onto the embedding space (see [51]). This decomposition relies on a combination of ecological proxies (*e.g.,* phylogentic relatedness [57]) in conjunction with known interactions to infer the latent values of species, which can then be mapped onto decomposition of a known network.

4.2.2 Species agnostic networks: construction through structure induction

These models generate ecologically realistic network structures using simple probabilistic rules. They are commonly used as null expectations, for comparative analyses, and as inputs to dynamical models. The determination of links between species is not directly linked to properties of the nodes. This means these networks are usually not species specific. Although they require little empirical information, they encode explicit assumptions about expected network structure.

Stochastic network models use a probabilistic rule-set about diet choice and niche breadth to reflect fundamental ideas of foraging biology. These models that are based on the compartmentalisation and acquisition of energy for species at different trophic levels [112,113] and that network structure can be determined by distributing interactions along single dimension [the ‘niche axis’; [114]]. Typically these models parametrise some aspect of the network structure [although see 112 for a parameter-free model]. These models include the most commonly used network generator, the Niche model [39], as well as the original Cascade model [5] and the derived Nested hierarchy model [115]. Despite the fact that these networks are derived without any real world data they are still able to recover the structure of emirical networks [116]. These models often form the basis for dynamic models *e.g.,* the allometric trophic network [24,26] and bioenergetic food web models [34].

4.3 Construction through deduction

In contrast to the above approaches centred on feasibility, realised networks via methods reflecting abundance and diet choice typically rely on deductive reasoning and have a unique agenda to those above. In contrast to the inductive methods, inference about a realised network follows from a set of premises defining generative processes, often referred to as mechanisms, allowing inference about the distribution and strength of interactions. These models are widely used to study environmental gradients, energy flux, extinction dynamics, and ecosystem functioning. They also provide the structural backbone for dynamical systems modelling to address questions about stability-structure-productivity-function relationships, secondary extinction dynamics, species invasion and climate change. There are two broad groups of models in this deductive category.

4.3.1 Species-specific networks

These models capture the behaviour of the nodes by explicitly taking into account the properties of the different species in the community. Which means that there is a degree of variance in which links are predicted between species unlike the more 'static' predictions made by inductive models. However, these networks are 'costly' to construct in real world settings (requiring data about the entire community, as it is the behaviour of the system that determines the behaviour of the part) and also lack the larger diet niche context afforded by metawebs.

Neutral networks are built on the assumption that foraging decisions are tied *only* to the abundance of species within the community [117,118]. Here links are solely determined by the relative abundance of the different species in the community. Although unrealistic as a complete explanation, neutral models can be combined with inductive approaches to generate more localised predictions [67].

There is a broader group of models that focus on determining interactions in terms of energetic constraints on diet breadth, often using the ratio of consumer-resource bodysize as a proxy for capturing the energetic constraints of feeding. Models such as those developed by [71] and [68] are similar to the mechanistic approaches discussed in Section 4.2, however instead of determining interactions based on mechanistic feasibility it is rather constrained by the energetic cost of predation. Note that although these models do not place any explicit constraints on the expected structure of the network, the links should still be considered as 'realised' owing to the energetic constraint placed on links. A different subset of diet models [*e.g.,* 42,119] use a diet choice approach, however similar to the stochastic network models they also embed assumptions on network structure. Thus these models predict both interactions and network structure simultaneously, although they would benefit in being refined by more explicitly accounting for trait-based (*i.e.,* feasibility) parameterisation [35].

5 Making Progress with Networks

The motivation to leverage network ecology in conservation ecology, environmental risk assessment and natural resource management stems from a shift away from species/population specific measures of the effects of stress and disturbance to community level metrics of these impacts. These metrics, such as resilience and more generally stability, ecosystem function and biodiversity *per se*, are natural properties of networks. This suggests that modern conservation, risk assessment and resource management requires robust network tools to support decision making.

This is also true in the disciplines of ecology and environmental science and their focus on abundance, distribution, functions and services that biodiversity provides [4]. Major questions remain, for example, about stability-diversity-productivity relationships, the impacts of extinctions and invasions and the impacts of multiple stressors operating at multiple ecological scales. A network approach to answering these types of questions specifically allows us to evaluate how environmental gradients and anthropogenic stress map through direct and indirect effects among species in a complex community and reveal fundamental patterns and understanding of processes in the natural world.

In order to effectively use networks to aid us in answering questions about conservation/risk assessment/management and core ecological theory, we need to be mindful that we are mapping the *correct* network representation to the question of interest [21]. Notably, there are certain questions that cannot be answered using specific network representations as the scale of the question of interest is fundamentally misaligned with either the process captured by a specific network representation Section 3.1, the underlying data that is used to construct it Section 4 or both of these factors.

Here we discuss and map the different network representations shown in Figure 1 to 'appropriate' research questions and agendas see also 1. We also highlight some of the key methodological challenges that currently limit our conceptualisation of a 'network' and thus impact their effective practical application in real world settings.

Table 1: Showcasing some of the broader avenues of inquiry, specifically how they map to the different network representations. Additionally we highlight some studies that address or present opening discussions around each research question. Superscripts at the research question indicate the strength of the current literature in addressing that research question: $^{\checkmark}$ indicates area with strong foundational research, $^{\Delta}$ partial/emerging areas of research, and $^{\times}$ areas where research is weak/largely absent

Network Representation	Example Research Question	Representative Studies
Global Metaweb	How will novel communities respond to *e.g.,* extinction, turnover, invasion and rewilding $^{\checkmark}$	[120]; [7]
	Diet-based conservation focusing not only on the target species but the species it might depend on for food resources $^{\Delta}$	[121]; [122]; [123]
	Rewiring capacity/potential of species by looking at the *entire* diets of species $^{\Delta}$	[124]; [31]; [125]; [126]
	Eco-Evolutionary dynamics and how they relate to the conservation and origination of feeding strategies $^{\times}$	[60]; [127]
Regional Metawebs	Applied use potential of questions highlighted for global metawebs at the management scale *e.g.,* a protected area $^{\Delta}$	[101]; [128]; [129]

Network Representation	Example Research Question	Representative Studies
	Refinement/extension of species distribution models by incorporating co-occurrence and species associations *e.g.*, predator and prey ✓	[130]; [131]
Realised webs	The allocation of multiple stressors across networks △	[132]; [133]
	Temperature threshold to community collapse △	[134]; [135]
	Extinction and persistence after harvesting/invasion/extinction ✓	[136]; [137]
	Stability-diversity-productivity-function ✓	[138]; [121]
	Explicilty tying ecosystem level processes and nutrient flows to networks ×	[139]
	Meta communities and the idea of meta-network-communities △	[140]; [141]

5.1 Key Eco-Evo-Conservation Questions

5.1.1 Global Metawebs

Global metawebs are most appropriate for questions centred on interaction feasibility and potential diet breadth. They provide a platform for exploring hypothetical or novel communities under climate change [142], species invasions, reintroductions, and rewilding, and the potential rewiring capacity [126]. Because they focus on feasible rather than realised interactions, global metawebs are also well suited to studying eco-evolutionary dynamics and how evolutionary history, natural selection, and phenotypic plasticity shape interaction niches.

5.1.2 Regional Metawebs

Regional metawebs extend these questions by explicitly incorporating species co-occurrence and therefore provide a more management-relevant perspective. They can be used to refine species distribution models and projections of future community composition [29,40]. However, caution is required when comparing regional metaweb structure across space or environmental gradients, as observed differences may reflect species turnover (*e.g.,* β-diversity) rather than changes in interaction processes.

5.1.3 Realised networks

Realised networks are best suited to questions concerning how community and environmental context shape interaction structure and ecosystem functioning [143]. They provide the appropriate framework for studying stability, resilience, biodiversity dynamics, ecosystem functioning, extinction cascades, invasions, climate change impacts, and network rewiring through time. By capturing interactions that are actually expressed, realised networks allow investigation of how perturbations propagate through communities and influence persistence.

The increasing availability of long-term interaction datasets is expanding opportunities to address these questions [27,144]. However, empirical datasets often accumulate interactions across extended periods, potentially obscuring temporal variation in realised interactions [10]. Developing approaches that better reconcile empirical networks with realised community dynamics therefore remains an important challenge.

5.2 Key methodological challenges

As noted above, the three network types highlight longstanding methodological challenges that limit both the precision and accuracy of ecological inference. Here we briefly review these challenges and emerging opportunities to address them.

Understanding what empirical data represents: Robust inference requires understanding what constitutes an observed interaction, whether recorded directly (predation events) or indirectly (*e.g.,* gut contents or stable isotopes). Because empirical networks often accumulate observations across space and time, they may be conceptually closer to metawebs than realised networks.

The validation of network structure: Considerable progress has been made in assessing how well models recover pairwise interactions [91,145], yet there remains no clear framework for evaluating their ability to recover network structure [114,146]. This raises two related questions: what constitutes an appropriate benchmark, and which aspects of network recovery matter most? For metawebs, accurately identifying both present and forbidden links may be essential, whereas for realised webs it remains unclear whether recovering pairwise interactions, aggregate properties (*e.g.,* connectance), or both should be the primary objective.

Transitioning between metawebs and realised webs: CMost approaches for modelling realised networks do not explicitly incorporate evolutionary constraints (although see [147]; [68]). Progress will likely require either ensemble approaches [148,149] or methods for downscaling metawebs into realised networks [e.g., 150]. However, any such transition must retain clarity about the meaning of links - structurally realistic networks do not necessarily represent realised prey choice or energy flow.

Developing frameworks that allow transitioning between metaweb and realised representations will also facilitate integration with metacommunity, metaecosystem, and ecosystem-level theory [151,152]. Doing so requires clear definitions of what constitutes a network and where its spatial and temporal boundaries lie [153].

Making networks more tractable in applied spaces: The application of networks to conservation and management remains limited by difficulties in defining network boundaries, aligning them with management units, and interpreting network metrics in policy-relevant ways [154]. Addressing these challenges will require stronger links between network structure and ecosystem function, alongside careful matching of analytical tools and network representations to management objectives [128,155].

Taken together, these challenges highlight three overarching messages. (i) Network representations are inseparable from the data and assumptions used to construct them; (ii) validation and benchmarking must be aligned with the intended network type and research question; and (iii) greater conceptual clarity is needed when transitioning between metawebs and realised networks, particularly in applied contexts. Explicitly articulating these distinctions is essential if networks are to be used rigorously and transparently across scales.

6 Conclusion

1. **Network representations encode different processes:** Different network representations capture distinct assumptions, data structures, and ecological processes; understanding the interplay between a network's structure and the processes it encodes is critical for determining which questions it can address.

2. **Network representations are not interchangeable:** There is no universally optimal network representation. The utility of a network depends entirely on how well its underlying assumptions and data align with the intended scale and process of analysis.

3. **Explicit assumptions are essential for evaluation:** Explicitly articulating the assumptions behind network assembly facilitates a more critical evaluation of network suitability and inferential power. Consequently, researchers can move beyond the uncritical use and default reliance on conventional network representations.

4. **A unified framework improves application and progress:** The framework presented here provides a structured basis for comparing network representations and evaluating their suitability across conceptual, methodological, and applied contexts. Establishing this standard is essential for preventing the misinterpretation of network data and for addressing foundational questions in biodiversity science.

References

1. Elton CS. 2001 *Animal Ecology*. Chicago, IL: University of Chicago Press. See https://press.uchicago.edu/ucp/books/book/chicago/A/bo25281897.html.

2. Lindeman RL. 1942 The Trophic-Dynamic Aspect of Ecology. *Ecology* **23**, 399–417. (doi:10.2307/1930126)

3. MacArthur R. 1955 Fluctuations of Animal Populations and a Measure of Community Stability. *Ecology* **36**, 533–536. (doi:10.2307/1929601)

4. Loreau M, de Mazancourt C. 2013 Biodiversity and ecosystem stability: A synthesis of underlying mechanisms. *Ecology Letters* **16**, 106–115. (doi:10.1111/ele.12073)

442 5. Cohen JE, Briand F, Newman C. 1990 *Community Food Webs: Data and Theory*. Berlin Heidelberg: Springer-Verlag. See https://www.springer.com/gp/book/9783642837869.

443 6. Martinez ND, Dunne JA. 1998 Time, space, and beyond: Scale issues in food-web research. In *Ecological Scale: Theory and Applications*, pp. 207–226. New York: Columbia University Press.

444 7. Dunne JA, Williams RJ, Martinez ND. 2002 Network structure and biodiversity loss in food webs: Robustness increases with connectance. *Ecology Letters* **5**, 558–567. (doi:10.1046/j.1461-0248.2002.00354.x)

445 8. Simmons BI *et al.* 2021 Refocusing multiple stressor research around the targets and scales of ecological impacts. *Nat Ecol Evol* **5**, 1478–1489. (doi:10.1038/s41559-021-01547-4)

446 9. Windsor FM, van den Hoogen J, Crowther TW, Evans DM. 2023 Using ecological networks to answer questions in global biogeography and ecology. *Journal of Biogeography* **50**, 57–69. (doi:10.1111/jbi.14447)

447 10. Polis GA. 1991 Complex Trophic Interactions in Deserts: An Empirical Critique of Food-Web Theory. *The American Naturalist* **138**, 123–155.

448 11. Martinez ND. 1991 Artifacts or Attributes? Effects of Resolution on the Little Rock Lake Food Web. *Ecological Monographs* **61**, 367–392. (doi:10.2307/2937047)

449 12. Dormann CF. 2023 The rise, and possible fall, of network ecology. In *Defining Agroecology – A Festschrift for Teja Tscharntke*, pp. 143–159. Hamburg: Tredition.

450 13. Blüthgen N. 2010 Why network analysis is often disconnected from community ecology: A critique and an ecologist's guide. *Basic and Applied Ecology* **11**, 185–195. (doi:10.1016/j.baae.2010.01.001)

451 14. Blüthgen N, Staab M. 2024 A Critical Evaluation of Network Approaches for Studying Species Interactions. *Annual Review of Ecology, Evolution, and Systematics* **55**, 65–88. (doi:10.1146/annurev-ecolsys-102722-021904)

452 15. Brimacombe C, Bodner K, Michalska-Smith M, Poisot T, Fortin M-J. 2023 Shortcomings of reusing species interaction networks created by different sets of researchers. *PLOS Biology* **21**, e3002068. (doi:10.1371/journal.pbio.3002068)

453 16. Brimacombe C, Bodner K, Gravel D, Leroux SJ, Poisot T, Fortin M-J. 2024 Publication-driven consistency in food web structures: Implications for comparative ecology. *Ecology* **n/a**, e4467. (doi:10.1002/ecy.4467)

454 17. Moulatlet G, Luna P, Dattilo W, Villalobos F. 2024 The scaling of trophic specialization in interaction networks across levels of organization. (doi:10.22541/au.172977303.33335171/v1)

455 18. Pringle RM, Hutchinson MC. 2020 Resolving Food-Web Structure. *Annual Review of Ecology, Evolution and Systematics* **51**, 55–80. (doi:10.1146/annurev-ecolsys-110218-024908)

456 19. Saberski E, Lorimer T, Carpenter D, Deyle E, Merz E, Park J, Pao GM, Sugihara G. 2024 The impact of data resolution on dynamic causal inference in multiscale ecological networks. *Commun Biol* **7**, 1–10. (doi:10.1038/s42003-024-07054-z)

457 20. Pimm SL. 1984 The complexity and stability of ecosystems. *Nature* **307**, 321–326. (doi:10.1038/307321a0)

458 21. Gauzens B *et al.* 2025 Tailoring interaction network types to answer different ecological questions. *Nat. Rev. Biodivers.*, 1–10. (doi:10.1038/s44358-025-00056-7)

459 22. Odum EP. 1968 Energy Flow in Ecosystems: A Historical Review. *Am Zool* **8**, 11–18. (doi:10.1093/icb/8.1.11)

23. May RM. 1972 Will a large complex system be stable? *Nature* **238**, 413–414. (doi:10.1038/238413a0)

24. Brose U, Williams RJ, Martinez ND. 2006 Allometric scaling enhances stability in complex food webs. *Ecology Letters* **9**, 1228–1236. (doi:10.1111/j.1461-0248.2006.00978.x)

25. Montoya JM, Pimm SL, Solé RV. 2006 Ecological networks and their fragility. *Nature* **442**, 259–264. (doi:10.1038/nature04927)

26. Schneider FD, Brose U, Rall BC, Guill C. 2016 Animal diversity and ecosystem functioning in dynamic food webs. *Nat Commun* **7**, 12718. (doi:10.1038/ncomms12718)

27. Danet A, Kéfi S, Johnson TF, Beckerman AP. 2024 Response diversity is a major driver of temporal stability in complex food webs., 2024.08.29.610288. (doi:10.1101/2024.08.29.610288)

28. Pilosof S, Porter MA, Pascual M, Kéfi S. 2017 The multilayer nature of ecological networks. *Nat Ecol Evol* **1**, 101. (doi:10.1038/s41559-017-0101)

29. Hao X, Holyoak M, Zhang Z, Yan C. 2025 Global Projection of Terrestrial Vertebrate Food Webs Under Future Climate and Land-Use Changes. *Global Change Biology* **31**, e70061. (doi:10.1111/gcb.70061)

30. Pecuchet L, Blanchet M-A, Frainer A, Husson B, Jørgensen LL, Kortsch S, Primicerio R. 2020 Novel feeding interactions amplify the impact of species redistribution on an Arctic food web. *Global Change Biology* **26**, 4894–4906. (doi:10.1111/gcb.15196)

31. Staniczenko PPA, Lewis OT, Jones NS, Reed-Tsochas F. 2010 Structural dynamics and robustness of food webs. *Ecology Letters* **13**, 891–899. (doi:10.1111/j.1461-0248.2010.01485.x)

32. Keyes AA, Barner AK, Dee LE. 2024 Synthesising the Relationships Between Food Web Structure and Robustness. *Ecology Letters* **27**, e14533. (doi:10.1111/ele.14533)

33. Keyes AA, McLaughlin JP, Barner AK, Dee LE. 2021 An ecological network approach to predict ecosystem service vulnerability to species losses. *Nat Commun* **12**, 1586. (doi:10.1038/s41467-021-21824-x)

34. Delmas E, Brose U, Gravel D, Stouffer DB, Poisot T. 2017 Simulations of biomass dynamics in community food webs. *Methods in Ecology and Evolution* **8**, 881–886. (doi:10.1111/2041-210X.12713)

35. Curtsdotter A, Banks HT, Banks JE, Jonsson M, Jonsson T, Laubmeier AN, Traugott M, Bommarco R. 2019 Ecosystem function in predator–prey food webs—confronting dynamic models with empirical data. *Journal of Animal Ecology* **88**, 196–210. (doi:10.1111/1365-2656.12892)

36. Terry JCD, Bonsall MB, Morris RJ. 2025 The impact of structured higher-order interactions on ecological network stability. *Theor Ecol* **18**, 9. (doi:10.1007/s12080-025-00603-0)

37. Poisot T, Stouffer DB, Kéfi S. 2016 Describe, understand and predict: Why do we need networks in ecology? *Functional Ecology* **30**, 1878–1882.

38. Yodzis P. 1982 The Compartmentation of Real and Assembled Ecosystems. *The American Naturalist* **120**, 551–570. (doi:10.1086/284013)

39. Williams RJ, Martinez ND. 2000 Simple rules yield complex food webs. *Nature* **404**, 180–183. (doi:10.1038/35004572)

40. García-Callejas D, Godoy O, Buche L, Hurtado M, Lanuza JB, Allen-Perkins A, Bartomeus I. 2023 Non-random interactions within and across guilds shape the potential to coexist in multi-trophic ecological communities. *Ecology Letters* **26**, 831–842. (doi:10.1111/ele.14206)

41. Clegg T, Ali M, Beckerman AP. 2018 The impact of intraspecific variation on food web structure. *Ecology* **99**, 2712–2720. (doi:10.1002/ecy.2523)

42. Beckerman AP, Petchey OL, Warren PH. 2006 Foraging biology predicts food web complexity. *Proc. Natl. Acad. Sci. U.S.A.* **103**, 13745–13749. (doi:10.1073/pnas.0603039103)

43. Dunhill AM, Zarzyczny K, Shaw JO, Atkinson JW, Little CTS, Beckerman AP. 2024 Extinction cascades, community collapse, and recovery across a Mesozoic hyperthermal event. *Nat Commun* **15**, 8599. (doi:10.1038/s41467-024-53000-2)

44. Dunne JA. 2006 The Network Structure of Food Webs. In *Ecological networks: Linking structure and dynamics* (eds JA Dunne, M Pascual), pp. 27–86. Oxford University Press.

45. Pringle RM. 2020 Untangling Food Webs. In *Unsolved Problems in Ecology*, pp. 225–238. Princeton University Press. (doi:10.1515/9780691195322-020)

46. Proulx SR, Promislow DEL, Phillips PC. 2005 Network thinking in ecology and evolution. *Trends in Ecology & Evolution* **20**, 345–353. (doi:10.1016/j.tree.2005.04.004)

47. Banville F *et al.* 2025 Deciphering Probabilistic Species Interaction Networks. *Ecology Letters* **28**, e70161. (doi:10.1111/ele.70161)

48. Poisot T, Cirtwill A, Cazelles K, Gravel D, Fortin M-J, Stouffer D. 2016 The structure of probabilistic networks. *Methods in Ecology and Evolution* **7**, 303–312. (doi:10)

49. Berlow EL *et al.* 2004 Interaction strengths in food webs: Issues and opportunities. *J Anim Ecology* **73**, 585–598. (doi:10.1111/j.0021-8790.2004.00833.x)

50. Jordano P. 2016 Sampling networks of ecological interactions. *Functional Ecology* (doi:10.1111/1365-2435.12763)

488 51. Strydom T *et al.* 2023 Graph embedding and transfer learning can help predict potential species interaction networks despite data limitations. *Methods in Ecology and Evolution* **14**, 2917–2930. (doi:10.1111/2041-210X.14228)

489 52. Segar ST, Fayle TM, Srivastava DS, Lewinsohn TM, Lewis OT, Novotny V, Kitching RL, Maunsell SC. 2020 The Role of Evolution in Shaping Ecological Networks. *Trends in Ecology & Evolution* **35**, 454–466. (doi:10.1016/j.tree.2020.01.004)

490 53. Gómez JM, Verdú M, Perfectti F. 2010 Ecological interactions are evolutionarily conserved across the entire tree of life. *Nature* **465**, 918–921. (doi:10.1038/nature09113)

491 54. Dalla Riva GV, Stouffer DB. 2016 Exploring the evolutionary signature of food webs' backbones using functional traits. *Oikos* **125**, 446–456. (doi:10.1111/oik.02305)

492 55. Rossberg AG, Matsuda H, Amemiya T, Itoh K. 2006 Food webs: Experts consuming families of experts. *Journal of Theoretical Biology* **241**, 552–563. (doi:10.1016/j.jtbi.2005.12.021)

493 56. Benadi G, Dormann CF, Fründ J, Stephan R, Vázquez DP. 2022 Quantitative Prediction of Interactions in Bipartite Networks Based on Traits, Abundances, and Phylogeny. *The American Naturalist* **199**, 841–854. (doi:10.1086/714420)

494 57. Strydom T *et al.* 2022 Food web reconstruction through phylogenetic transfer of low-rank network representation. *Methods in Ecology and Evolution* **13**, 2838–2849. (doi:10.1111/2041-210X.13835)

495 58. Blanchet FG, Cazelles K, Gravel D. 2020 Co-occurrence is not evidence of ecological interactions. *Ecology Letters* **23**, 1050–1063. (doi:10.1111/ele.13525)

496 59. Dansereau G, Barros C, Poisot T. 2024 Spatially explicit predictions of food web structure from regional-level data. *Philosophical Transactions of the Royal Society B: Biological Sciences* **379**. (doi:10.1098/rstb.2023.0166)

497 60. Poisot T, Stouffer DB, Gravel D. 2015 Beyond species: Why ecological interaction networks vary through space and time. *Oikos* **124**, 243–251. (doi:10.1111/oik.01719)

498 61. Caron D, Brose U, Lurgi M, Blanchet FG, Gravel D, Pollock LJ. 2024 Trait-matching models predict pairwise interactions across regions, not food web properties. *Global Ecology and Biogeography* **33**, e13807. (doi:10.1111/geb.13807)

499 62. Quintero E, Arroyo-Correa B, Isla J, Rodríguez-Sánchez F, Jordano P. 2024 Downscaling mutualistic networks from species to individuals reveals consistent interaction niches and roles within plant populations., 2024.02.02.578595. (doi:10.1101/2024.02.02.578595)

500 63. Stephens DW, Krebs JR. 1986 *Foraging Theory*. Princeton University Press. (doi:10.2307/j.ctvs32s6b)

501 64. Vázquez DP, Blüthgen N, Cagnolo L, Chacoff NP. 2009 Uniting pattern and process in plant–animal mutualistic networks: A review. *Annals of Botany* **103**, 1445–1457. (doi:10.1093/aob/mcp057)

502 65. Canard E, Mouquet N, Marescot L, Gaston KJ, Gravel D, Mouillot D. 2012 Emergence of Structural Patterns in Neutral Trophic Networks. *PLOS ONE* **7**, e38295. (doi:10.1371/journal.pone.0038295)

503 66. Momal R, Robin S, Ambroise C. 2020 Tree-based inference of species interaction networks from abundance data. *Methods in Ecology and Evolution* **11**, 621–632. (doi:10.1111/2041-210X.13380)

504 67. Pomeranz JPF, Thompson RM, Poisot T, Harding JS. 2019 Inferring predator–prey interactions in food webs. *Methods in Ecology and Evolution* **10**, 356–367. (doi:10.1111/2041-210X.13125)

68. Wootton KL, Curtsdotter A, Roslin T, Bommarco R, Jonsson T. 2023 Towards a modular theory of trophic interactions. *Functional Ecology* **37**, 26–43. (doi:10.1111/1365-2435.13954)

69. Smith JG, Tomoleoni J, Staedler M, Lyon S, Fujii J, Tinker MT. 2021 Behavioral responses across a mosaic of ecosystem states restructure a sea otter–urchin trophic cascade. *Proceedings of the National Academy of Sciences* **118**, e2012493118. (doi:10.1073/pnas.2012493118)

70. Pawar S, Dell AI, Savage VM. 2012 Dimensionality of consumer search space drives trophic interaction strengths. *Nature* **486**, 485–489. (doi:10.1038/nature11131)

71. Portalier SMJ, Fussmann GF, Loreau M, Cherif M. 2019 The mechanics of predator–prey interactions: First principles of physics predict predator–prey size ratios. *Functional Ecology* **33**, 323–334. (doi:10.1111/1365-2435.13254)

72. Cherif M *et al.* 2024 The environment to the rescue: Can physics help predict predator–prey interactions? *Biological Reviews* **138**. (doi:10.1111/brv.13105)

73. Pyke G. 1984 Optimal Foraging Theory: A Critical Review. *Annual Review of Ecology, Evolution and Systematic* **15**, 523–575. (doi:10.1146/annurev.ecolsys.15.1.523)

74. Brown JH, Gillooly JF, Allen AP, Savage VM, West GB. 2004 Toward a Metabolic Theory of Ecology. *Ecology* **85**, 1771–1789. (doi:10.1890/03-9000)

75. Miele V, Guill C, Ramos-Jiliberto R, Kéfi S. 2019 Non-trophic interactions strengthen the diversity—functioning relationship in an ecological bioenergetic network model. *PLOS Computational Biology* **15**, e1007269. (doi:10.1371/journal.pcbi.1007269)

76. Ings TC *et al.* 2009 Ecological networks–beyond food webs. *J Anim Ecol* **78**, 253–269. (doi:10.1111/j.1365-2656.2008.01460.x)

514 77. Golubski AJ, Abrams PA. 2011 Modifying modifiers: What happens when interspecific interactions interact? *Journal of Animal Ecology* **80**, 1097–1108. (doi:10.1111/j.1365-2656.2011.01852.x)

515 78. Kamaru DN *et al.* 2024 Disruption of an ant-plant mutualism shapes interactions between lions and their primary prey. *Science* **383**, 433–438. (doi:10.1126/science.adg1464)

516 79. Holland JN, DeAngelis DL, Bronstein JL. 2002 Population Dynamics and Mutualism: Functional Responses of Benefits and Costs. *The American Naturalist* **159**, 231–244. (doi:10.1086/338510)

517 80. Bascompte J, Jordano P. 2007 Plant-Animal Mutualistic Networks: The Architecture of Biodiversity. *Annual Review of Ecology, Evolution, and Systematics* **38**, 567–593. (doi:10.1146/annurev.ecolsys.38.091206.095818)

518 81. Sauve AMC, Thébault E, Pocock MJO, Fontaine C. 2016 How plants connect pollination and herbivory networks and their contribution to community stability. *Ecology* **97**, 908–917. (doi:10.1890/15-0132.1)

519 82. Kéfi S *et al.* 2012 More than a meal… integrating non-feeding interactions into food webs. *Ecology Letters* **15**, 291–300. (doi:10.1111/j.1461-0248.2011.01732.x)

520 83. Kéfi S, Berlow EL, Wieters EA, Joppa LN, Wood SA, Brose U, Navarrete SA. 2015 Network structure beyond food webs: Mapping non-trophic and trophic interactions on Chilean rocky shores. *Ecology* **96**, 291–303. (doi:10.1890/13-1424.1)

521 84. Buche L, Bartomeus I, Godoy O. 2024 Multitrophic Higher-Order Interactions Modulate Species Persistence. *The American Naturalist* **203**, 458–472. (doi:10.1086/729222)

522 85. Jordano P. 2016 Chasing Ecological Interactions. *PLOS Biology* **14**, e1002559. (doi:10.1371/journal.pbio.1002559)

86. Poisot T, Bergeron G, Cazelles K, Dallas T, Gravel D, MacDonald A, Mercier B, Violet C, Vissault S. 2021 Global knowledge gaps in species interaction networks data. *Journal of Biogeography* **48**, 1552–1563. (doi:10.1111/jbi.14127)

87. Biton B, Puzis R, Pilosof S. 2024 Inductive link prediction boosts data availability and enables cross-community link prediction in ecological networks. See https://ecoevorxiv.org/repository/view/7492/ (accessed on 16 September 2024).

88. Stock M, Poisot T, Waegeman W, Baets BD. 2017 Linear filtering reveals false negatives in species interaction data. *Scientific Reports* **7**, 45908. (doi:10.1038/srep45908)

89. Dallas T, Park AW, Drake JM. 2017 Predicting cryptic links in host-parasite networks. *PLOS Computational Biology* **13**, e1005557. (doi:10.1371/journal.pcbi.1005557)

90. Poisot T *et al.* 2023 Network embedding unveils the hidden interactions in the mammalian virome. *Patterns* **4**, 100738. (doi:10.1016/j.patter.2023.100738)

91. Strydom T *et al.* 2021 A roadmap towards predicting species interaction networks (across space and time). *Philosophical Transactions of the Royal Society B: Biological Sciences* **376**, 20210063. (doi:10.1098/rstb.2021.0063)

92. Stouffer DB. 2019 All ecological models are wrong, but some are useful. *Journal of Animal Ecology* **88**, 192–195. (doi:10.1111/1365-2656.12949)

93. Morales-Castilla I, Matias MG, Gravel D, Araújo MB. 2015 Inferring biotic interactions from proxies. *Trends in Ecology & Evolution* **30**, 347–356. (doi:10.1016/j.tree.2015.03.014)

94. Bramon Mora B, Gravel D, Gilarranz LJ, Poisot T, Stouffer DB. 2018 Identifying a common backbone of interactions underlying food webs from different ecosystems. *Nat Commun* **9**, 2603. (doi:10.1038/s41467-018-05056-0)

95. Yeakel JD, Pires MM, Rudolf L, Dominy NJ, Koch PL, Guimarães PR, Gross T. 2014 Collapse of an ecological network in Ancient Egypt. *PNAS* **111**, 14472–14477. (doi:10.1073/pnas.1408471111)

96. Fricke EC, Hsieh C, Middleton O, Gorczynski D, Cappello CD, Sanisidro O, Rowan J, Svenning J-C, Beaudrot L. 2022 Collapse of terrestrial mammal food webs since the Late Pleistocene. *Science* **377**, 1008–1011. (doi:10.1126/science.abn4012)

97. Van der Putten WH, Macel M, Visser ME. 2010 Predicting species distribution and abundance responses to climate change: Why it is essential to include biotic interactions across trophic levels. *Philosophical Transactions of the Royal Society B: Biological Sciences* **365**, 2025–2034. (doi:10.1098/rstb.2010.0037)

98. Higino GT, Banville F, Dansereau G, Muñoz NRF, Windsor F, Poisot T. 2023 Mismatch between IUCN range maps and species interactions data illustrated using the Serengeti food web. *PeerJ* **11**, e14620. (doi:10.7717/peerj.14620)

99. Pollock LJ, Tingley R, Morris WK, Golding N, O'Hara RB, Parris KM, Vesk PA, McCarthy MA. 2014 Understanding co-occurrence by modelling species simultaneously with a Joint Species Distribution Model (JSDM). *Methods in Ecology and Evolution* **5**, 397–406. (doi:10.1111/2041-210X.12180)

100. Gravel D *et al.* 2019 Bringing Elton and Grinnell together: A quantitative framework to represent the biogeography of ecological interaction networks. *Ecography* **42**, 401–415. (doi:10.1111/ecog.04006)

101. Albouy C, Velez L, Coll M, Colloca F, Le Loc'h F, Mouillot D, Gravel D. 2014 From projected species distribution to food-web structure under climate change. *Global Change Biology* **20**, 730–741. (doi:10.1111/gcb.12467)

102. Shaw JO, Dunhill AM, Beckerman AP, Dunne JA, Hull PM. 2024 A framework for reconstructing ancient food webs using functional trait data., 2024.01.30.578036. (doi:10.1101/2024.01.30.578036)

540 103. Roopnarine PD. 2017 Ecological Modelling of Paleocommunity Food Webs. In *Conservation Paleobiology: Using the Past to Manage for the Future*, pp. 201–226. University of Chicago Press.

541 104. Gravel D, Poisot T, Albouy C, Velez L, Mouillot D. 2013 Inferring food web structure from predator–prey body size relationships. *Methods in Ecology and Evolution* **4**, 1083–1090. (doi:10.1111/2041-210X.12103)

542 105. Rohr RP, Scherer H, Kehrli P, Mazza C, Bersier L. 2010 Modeling Food Webs: Exploring Unexplained Structure Using Latent Traits. *The American Naturalist* **176**, 170–177. (doi:10.1086/653667)

543 106. Pichler M, Boreux V, Klein A-M, Schleuning M, Hartig F. 2020 Machine learning algorithms to infer trait-matching and predict species interactions in ecological networks. *Methods in Ecology and Evolution* **11**, 281–293. (doi:10.1111/2041-210X.13329)

544 107. Caron D, Maiorano L, Thuiller W, Pollock LJ. 2022 Addressing the Eltonian shortfall with trait-based interaction models. *Ecology Letters* **25**, 889–899. (doi:10.1111/ele.13966)

545 108. Llewelyn J *et al.* 2023 Predicting predator–prey interactions in terrestrial endotherms using random forest. *Ecography* **2023**, e06619. (doi:10.1111/ecog.06619)

546 109. Desjardins-Proulx P, Laigle I, Poisot T, Gravel D. 2017 Ecological interactions and the Netflix problem. *PeerJ* **5**, e3644. (doi:10.7717/peerj.3644)

547 110. Eklöf A, Tang S, Allesina S. 2013 Secondary extinctions in food webs: A Bayesian network approach. *Methods Ecol Evol* **4**, 760–770. (doi:10.1111/2041-210X.12062)

548 111. Cirtwill AR, Ekl f A, Roslin T, Wootton K, Gravel D. 2019 A quantitative framework for investigating the reliability of empirical network construction. *Methods in Ecology and Evolution* **10**, 902–911. (doi:10.1111/2041-210X.13180)

549 112. Allesina S, Pascual M. 2009 Food web models: A plea for groups. *Ecology Letters* **12**, 652–662. (doi:10.1111/j.1461-0248.2009.01321.x)

550 113. Krause AE, Frank KA, Mason DM, Ulanowicz RE, Taylor WW. 2003 Compartments revealed in food-web structure. *Nature* **426**, 282–285. (doi:10.1038/nature02115)

551 114. Allesina S, Alonso D, Pascual M. 2008 A General Model for Food Web Structure. *Science* **320**, 658–661. (doi:10.1126/science.1156269)

552 115. Cattin M-F, Bersier L-F, Banašek-Richter C, Baltensperger R, Gabriel J-P. 2004 Phylogenetic constraints and adaptation explain food-web structure. *Nature* **427**, 835–839. (doi:10.1038/nature02327)

553 116. Stouffer DB, Camacho J, Guimerà R, Ng CA, Nunes Amaral LA. 2005 Quantitative Patterns in the Structure of Model and Empirical Food Webs. *Ecology* **86**, 1301–1311. (doi:10.1890/04-0957)

554 117. Canard EF, Mouquet N, Mouillot D, Stanko M, Miklisova D, Gravel D. 2014 Empirical Evaluation of Neutral Interactions in Host-Parasite Networks. *The American Naturalist* **183**, 468–479. (doi:10.1086/675363)

555 118. Krishna A, Guimarães Jr PR, Jordano P, Bascompte J. 2008 A neutral-niche theory of nestedness in mutualistic networks. *Oikos* **117**, 1609–1618. (doi:10.1111/j.1600-0706.2008.16540.x)

556 119. Petchey OL, Beckerman AP, Riede JO, Warren PH. 2008 Size, foraging, and food web structure. *Proc. Natl. Acad. Sci. U.S.A.* **105**, 4191–4196. (doi:10.1073/pnas.0710672105)

557 120. Gravel D, Albouy C, Thuiller W. 2016 The meaning of functional trait composition of food webs for ecosystem functioning. *Philosophical Transactions: Biological Sciences* **371**, 1–14.

121. Rooney N, McCann KS. 2012 Integrating food web diversity, structure and stability. *Trends in Ecology & Evolution* **27**, 40–46. (doi:10.1016/j.tree.2011.09.001)

122. Curtsdotter A, Binzer A, Brose U, De Castro F, Ebenman B, Eklöf A, Riede JO, Thierry A, Rall BC. 2011 Robustness to secondary extinctions: Comparing trait-based sequential deletions in static and dynamic food webs. *Basic and Applied Ecology* **12**, 571–580. (doi:10.1016/j.baae.2011.09.008)

123. McDonald-Madden E, Sabbadin R, Game ET, Baxter PWJ, Chadès I, Possingham HP. 2016 Using food-web theory to conserve ecosystems. *Nat Commun* **7**, 10245. (doi:10.1038/ncomms10245)

124. Gilljam D, Curtsdotter A, Ebenman B. 2015 Adaptive rewiring aggravates the effects of species loss in ecosystems. *Nat Commun* **6**, 8412. (doi:10.1038/ncomms9412)

125. Su M, Ma Q, Hui C. 2024 Adaptive rewiring shapes structure and stability in a three-guild herbivore-plant-pollinator network. *Commun Biol* **7**, 103. (doi:10.1038/s42003-024-05784-8)

126. Marjakangas E-L, Dalsgaard B, Ordonez A. 2025 Fundamental Interaction Niches: Towards a Functional Understanding of Ecological Networks' Resilience. *Ecology Letters* **28**, e70146. (doi:10.1111/ele.70146)

127. Baskerville EB, Dobson AP, Bedford T, Allesina S, Anderson TM, Pascual M. 2011 Spatial Guilds in the Serengeti Food Web Revealed by a Bayesian Group Model. *PLOS Computational Biology* **7**, e1002321. (doi:10.1371/journal.pcbi.1002321)

128. Pellissier L *et al.* 2018 Comparing species interaction networks along environmental gradients. *Biological Reviews* **93**, 785–800. (doi:10.1111/brv.12366)

129. Estrada E, Bodin Ö. 2008 Using Network Centrality Measures to Manage Landscape Connectivity. *Ecological Applications* **18**, 1810–1825. (doi:10.1890/07-1419.1)

567 130. Araújo MB, Luoto M. 2007 The importance of biotic interactions for modelling species distributions under climate change. *Global Ecol Biogeography* **16**, 743–753. (doi:10.1111/j.1466-8238.2007.00359.x)

568 131. Kissling WD *et al.* 2012 Towards novel approaches to modelling biotic interactions in multispecies assemblages at large spatial extents. *Journal of Biogeography* **39**, 2163–2178. (doi:10.1111/j.1365-2699.2011.02663.x)

569 132. Crain CM, Kroeker K, Halpern BS. 2008 Interactive and cumulative effects of multiple human stressors in marine systems. *Ecol Lett* **11**, 1304–1315. (doi:10.1111/j.1461-0248.2008.01253.x)

570 133. Beauchesne D, Cazelles K, Archambault P, Dee LE, Gravel D. 2021 On the sensitivity of food webs to multiple stressors. *Ecol Lett* **24**, 2219–2237. (doi:10.1111/ele.13841)

571 134. O'Gorman EJ *et al.* 2019 A simple model predicts how warming simplifies wild food webs. *Nat. Clim. Chang.* **9**, 611–616. (doi:10.1038/s41558-019-0513-x)

572 135. Petchey OL, Brose U, Rall BC. 2010 Predicting the effects of temperature on food web connectance. *Philosophical Transactions of the Royal Society B: Biological Sciences* **365**, 2081–2091. (doi:10.1098/rstb.2010.0011)

573 136. Allesina S, Tang S. 2012 Stability criteria for complex ecosystems. *Nature* **483**, 205–208. (doi:10.1038/nature10832)

574 137. Yodzis P. 2001 Must top predators be culled for the sake of fisheries? *Trends in Ecology & Evolution* **16**, 78–84. (doi:10.1016/S0169-5347(00)02062-0)

575 138. Thébault E, Fontaine C. 2010 Stability of ecological communities and the architecture of mutualistic and trophic networks. *Science* **329**, 853–856. (doi:10.1126/science.1188321)

576 139. Moore JC *et al.* 2004 Detritus, trophic dynamics and biodiversity. *Ecology Letters* **7**, 584–600. (doi:10.1111/j.1461-0248.2004.00606.x)

577 140. Gravel D, Massol F, Leibold MA. 2016 Stability and complexity in model meta-ecosystems. *Nature Communications* **7**, 12457. (doi:10.1038/ncomms12457)

578 141. Gilarranz LJ, Rayfield B, Liñán-Cembrano G, Bascompte J, Gonzalez A. 2017 Effects of network modularity on the spread of perturbation impact in experimental metapopulations. *Science* **357**, 199–201. (doi:10.1126/science.aal4122)

579 142. Hui C, Richardson DM. 2019 How to Invade an Ecological Network. *Trends in Ecology & Evolution* **34**, 121–131. (doi:10.1016/j.tree.2018.11.003)

580 143. Thuiller W, Calderón-Sanou I, Chalmandrier L, Gaüzère P, O'Connor LMJ, Ohlmann M, Poggiato G, Münkemüller T. 2024 Navigating the integration of biotic interactions in biogeography. *Journal of Biogeography* **51**, 550–559. (doi:10.1111/jbi.14734)

581 144. Wooster EIF *et al.* 2024 Australia's recently established predators restore complexity to food webs simplified by extinction. *Current Biology* **34**, 5164–5172.e2. (doi:10.1016/j.cub.2024.09.049)

582 145. Poisot T. 2023 Guidelines for the prediction of species interactions through binary classification. *Methods in Ecology and Evolution* **14**, 1333–1345. (doi:10.1111/2041-210X.14071)

583 146. Tylianakis JM, Laliberté E, Nielsen A, Bascompte J. 2010 Conservation of species interaction networks. *Biological Conservation* **143**, 2270–2279. (doi:10.1016/j.biocon.2009.12.004)

584 147. Van De Walle R, Logghe G, Haas N, Massol F, Vandegehuchte ML, Bonte D. 2023 Arthropod food webs predicted from body size ratios are improved by incorporating prey defensive properties. *Journal of Animal Ecology* **92**, 913–924. (doi:10.1111/1365-2656.13905)

585 148. Becker DJ *et al.* 2022 Optimising predictive models to prioritise viral discovery in zoonotic reservoirs. *The Lancet Microbe* **3**, e625–e637. (doi:10.1016/S2666-5247(21)00245-7)

586 149. Terry JCD, Lewis OT. 2020 Finding missing links in interaction networks. *Ecology* **101**, e03047. (doi:10.1002/ecy.3047)

587 150. Roopnarine PD. 2006 Extinction Cascades and Catastrophe in Ancient Food Webs. *Paleobiology* **32**, 1–19.

588 151. Massol F, Gravel D, Mouquet N, Cadotte MW, Fukami T, Leibold MA. 2011 Linking community and ecosystem dynamics through spatial ecology. *Ecology Letters* **14**, 313–323. (doi:10.1111/j.1461-0248.2011.01588.x)

589 152. Liu J, Liu M, Yin J, Zhang Y, Guo Y, Yu T, Yu H, Ming X. 2025 Harnessing food webs as a framework for enhancing ecosystem multifunctionality. *Biological Conservation* **310**, 111383. (doi:10.1016/j.biocon.2025.111383)

590 153. Fortin M-J, Dale MRT, Brimacombe C. 2021 Network ecology in dynamic landscapes. *Proc. R. Soc. B.* **288**, rspb.2020.1889, 20201889. (doi:10.1098/rspb.2020.1889)

591 154. Dansereau G *et al.* 2024 Overcoming the disconnect between interaction networks and biodiversity conservation and management.

592 155. O'Connor LMJ *et al.* 2025 The untapped potential of food webs in systematic conservation planning.

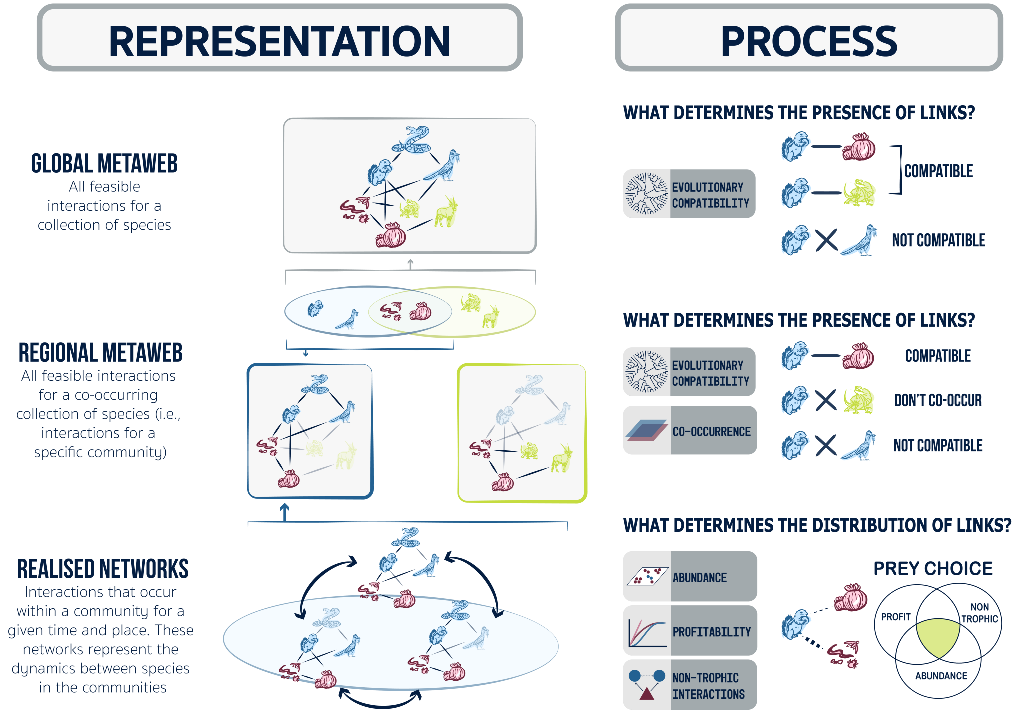

Figure 1: Aligning the processes that determine interactions (right) with different network representations (left). A **global metaweb** captures all feasible interactions among a species pool. Restricting this network by species co-occurrence yields **regional metawebs**, which differ in species composition and interaction feasibility. Species occurring in both regions are shown in red, whereas region-specific species are shown in blue and yellow. **Realised networks** represent subsets of regional metawebs expressed in particular spatial and temporal contexts. Their structure is shaped not only by co-occurrence, but also by community-level processes such as abundance, diet choice, and non-trophic interactions.